

Week 4 pre-read.

Name _____ Date _____

About 30 minutes. Graded complete or incomplete. Do this before the first pass.

This is a quick look ahead, not a full read. Look at the headings and the figures in both chapters, answer the questions below in a few words, and move on. You will learn all of it properly in the slides, and it will make much more sense because you have already seen it once.

Where to look

First, Silverthorn Chapter 5, Membrane Dynamics: the part near the end of the chapter on the resting membrane potential. You read the rest of Chapter 5 in Week 2, so this is the one piece you have not seen yet.

Then, Silverthorn Chapter 9, Neurons: Cellular and Network Properties. Skim the whole chapter. Read the headings and look at the figures, especially the ones named below.

Chapter numbers are for the 9th edition. Other editions number things differently, so go by the chapter title and the topic, and use the search in your eText to find it.

What this reading should help you do

Read each goal and pick the answer that fits you right now. There is no wrong answer here.

The resting membrane potential (Chapter 5)

Goal	New to me	Heard of it	Could explain
Explain what it means for a cell to have a resting membrane potential.	☐	☐	☐
Describe which ions create the resting membrane potential, and why.	☐	☐	☐
Explain how changing the membrane's permeability to an ion changes the membrane potential, with examples.	☐	☐	☐

Neurons (Chapter 9)

Goal	New to me	Heard of it	Could explain
Map how the nervous system is organized.	☐	☐	☐
Name the parts of a neuron and a synapse and what each part does, and name the types of glial cells.	☐	☐	☐
Explain how graded potentials and action potentials differ, which ions move during an action potential, what the refractory periods are, and how myelin speeds up conduction.	☐	☐	☐
Tell electrical synapses from chemical synapses, and describe how neurotransmitters are made, released, and stopped.	☐	☐	☐
Explain how a neuron adds up the excitatory and inhibitory signals it receives, and how a synapse can get stronger with use.	☐	☐	☐

Find it

Each answer is in the figure or page named. A few words is enough.

The resting membrane potential (Chapter 5)

Which positive ion is most concentrated inside the cell? Which one is most concentrated outside the cell?

Look at the bar graph in the Essentials figure called Body Fluid Compartments.

When a potassium leak channel is added to the artificial cell, which way does potassium move? Why can't the large negative ions follow it?

Look at the Essentials figure called Membrane Potential, the steps with the artificial cell.

Potassium eventually stops leaving the cell. What pulls it back in? Using the table of values in the figure, what is the equilibrium potential for potassium?

Same Essentials figure, the parts on electrochemical equilibrium and equilibrium potential.

Where are the two electrodes placed, and what value is the bath given? On the graph, what is it called when the line moves up (less negative), moves down (more negative), and comes back to rest?

Look at the figure called Measuring membrane potential.

Neurons (Chapter 9)

Name two ways a graded potential is different from an action potential.

Look at the table that compares graded potentials and action potentials.

During an action potential, which ion enters the cell in the rising phase, and which ion leaves in the falling phase? About what voltage is threshold on the graph?

Look at the Essentials figure called The Action Potential.

What does high blood potassium do to a neuron's resting membrane potential? What does low blood potassium do? You will use this with your patient this week.

Look at the figure called Potassium and cell excitability.

When an action potential reaches the axon terminal, which ion enters the terminal and triggers the release of neurotransmitter? Name one way the neurotransmitter's action is stopped afterward.

Look at the Essentials figure called Synaptic communication.

What is the difference between temporal summation and spatial summation? In the part on synaptic inhibition, why does the neuron not fire an action potential?

Look at the Essentials figure called Integration of Synaptic Signaling.

Words to know

Find each term in the reading. Write what it means in your own words, not the book's. If you are working on paper, add a quick sketch. If you are typing, give one example instead.

Resting membrane potential

Equilibrium potential

Depolarize

Hyperpolarize

Graded potential

Action potential

Myelin

Neurotransmitter

Before you start the slides

Chapter 9 opens with a short list of background topics it expects you to know already. Which one do you feel least sure about?

Write one question you want the slides to answer.

Turn it in: photograph or scan these pages and upload them to the Week 4 pre-read assignment in Canvas.